KONERU LAKSHMAIAH EDUCATION FOUNDATION
(Deemed to be University, Estd. u/s. 3 of UGC Act 1956)

B.Tech - Even Sem : Semester in Exam-I
Academic Year:2024-2025

23MT2004 - MATHEMATICAL PROGRAMMING

Set No: 3

Time: **Max.Marks: 50**

S.NO	Answer All Questions	Choice	Options	Marks	CO	CO BTL	COI BTL
1.	Answer the following.			12Marks	CO2	3	2
1.A.	Give the difference between a feasible and an optimal solution?			2Marks	CO1	3	2
1.B.	Explain why Karmarkar's Algorithm used for?			2Marks	CO1	3	2
1.C.	State four applications of Linear Programming.			2Marks	CO1	3	2
1.D.	Give the mathematical model of 0-1 knapsack problem.			2Marks	CO2	3	2
1.E.	State briefly about different types of Integer programming problems.			2Marks	CO2	3	2
1.F.	Explain where cutting plane method is used.			2Marks	CO2	3	2
2.	Answer the following.			16Marks	CO2	3	3
2.A.	A garment manufacturer has a production line making two styles of shirts. Style I needs 200 g of cotton thread, 300 g of Dacron thread and 300 g of linen thread. Corresponding requirements of style II are 200g, 200g and 100g. The net contributions are Rs.19.50 for style I and Rs.15.90 for style II. The available inventory of cotton thread, Dacron thread and linen thread are, respectively, 24 kg, 26 kg and 22 kg. The manufacturer wants to determine the number of each style to be produced with the given inventory. Formulate the LPP model.			4Marks	CO1	3	3
2.B.	Construct the dual of the problem $\text{maximize} \quad Z = 3x_1 + 10x_2 + 2x_3,$ $\text{subject to} \quad 2x_1 + 3x_2 + 2x_3 \le 7,$ $3x_1 - 2x_2 + 4x_3 = 3,$ $x_1 \ge 0,\ x_2 \ge 0,\ x_3 \ge 0.$			4Marks	CO1	3	3
2.C.	Convert the below fractional programming to Linear programming. $\text{Maximize} \quad Z(x) = \dfrac{5x_1 + 3x_2}{5x_1 + 2x_2 + 1},$ $\text{subject to} \quad 3x_1 + 5x_2 \le 15,$ $5x_1 + 2x_2 \le 10,$ $x_1, x_2 \ge 0.$			4Marks	CO2	3	3
2.D.	A thief enters a shop and finds 5 items with the following properties: The thief can carry a maximum weight of 8 units. Determine the maximum value he can steal.			4Marks	CO2	3	3

Item	Weight	Value
1	1	2
2	3	5
3	4	6
4	5	7
5	7	10

3.	Answer the following.	choice Q-4		11Marks	CO1	3	3
3.A.	Aditi holds two part-time jobs, Job I and Job II. She never wants to work more than a total of 12 hours a week. She has determined that for every hour she works at Job I, she needs 2 hours of preparation time, and for every hour she works at Job II, she needs one hour of preparation time, and she cannot spend more than 16 hours for preparation. If she makes $40 an hour at Job I, and $30 an hour at Job II, how many hours should she work per week at each job to maximize her income? First write this in Linear programming format.			5Marks	CO1	3	3
3.B.	Solve the above LP problem using simplex method.			6Marks	CO1	3	3
4.	Solve the following.			11Marks	CO1	3	3
4.A.	Find the Initial basic feasible solution for the below transportation problem.			5Marks	CO1	3	3

Destination

Source	D1	D2	D3	D4	D5	Supply (Si)
O1	5	1	8	7	5	15
O2	3	9	6	7	8	25
O3	4	2	7	6	5	42
O4	7	11	10	4	9	35
Demand (di)	30	20	15	10	20	117 / 95

4.B.	Find the optimal solution using MODI method for the above unbalanced transportation problem.			6Marks	CO1	3	3
5.	Answer the following.	choice Q-6		11Marks	CO2	3	3
5.A.	Solve the below Integer programming problem using branch and bound			5Marks	CO2	3	3

	maximize y = 5x1 + 8x2 Subject to 5x1 + 9x2 <= 45 x1 + x2 <= 6 x1 and x2 >= 0 **x1 and x2 are integers**						
5.B.	Give the tree of optimal solutions of all subproblems and give the best solution of above Integer programming once all the branches are fathomed.			6Marks	CO2	3	3
6.	Answer the following.			11Marks	CO2	3	3
6.A.	A delivery truck needs to visit 4 cities (A, B, C, D) and return to the starting city. The cost of traveling between the cities is as shown below. Solve this TSP using brute force search method.			5Marks	CO2	3	3

	A	B	C	D
A	0	8	6	12
B	8	0	10	15
C	6	10	0	9
D	12	15	9	0

6.B.	Solve the above TSP using dynamic programming.			6Marks	CO2	3	3

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